

WEISHUN ZHONG

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Princeton, New Jersey, 08540

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Research interests: The intersection of computational neuroscience, AI, and statistical physics.

EMPLOYMENT

Institute for Advanced Study, Princeton, NJ

Sep.2023-present

Postdoctoral Member, Simons Center for Systems Biology, School of Natural Sciences

Advisor: Misha Tsodyks

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA

Sep.2017-Jun.2023

Ph.D. Department of Physics

Advisors: Haim Sompolinsky (Harvard) and Mehran Kardar (MIT)

Thesis: Non-equilibrium Physics: from Spin Glasses to Machine and Neural Learning

University of Chicago, Chicago, IL

Sep.2016-Jun.2017

M.S., *Physical Sciences Division*, Physics

GPA: 3.93/4.0

Advisors: Arvind Murugan and David J. Schwab

University of Michigan, Ann Arbor, MI

Sep.2013-May.2016

B.S., *highest distinction*, Physics and Mathematics

GPA: 3.97/4.0

Advisor: James T. Liu

PUBLICATIONS

1. “[Semantic Chunking and the Entropy of Natural Language](#)”, **Weishun Zhong**, Doron Sivan, Tankut Can, Mikhail Katkov, Misha Tsodyks, *arXiv: 2602.13194*.
2. “[Random Tree Model of Meaningful Memory](#)”, **Weishun Zhong**, Tankut Can, Atonis Georgiou, Ilya Shnayderman, Mikhail Katkov, Misha Tsodyks, *arXiv: 2412.01806; Physical Review Letters 134.23 (2025): 237402*.
3. “[Synaptic Theory of Chunking in Working Memory](#)”, **Weishun Zhong**, Mikhail Katkov, Misha Tsodyks, *arXiv: 2408.07637, accepted at eLife*.
4. “[Advantage of Quantum Neural Networks as Quantum Information Decoders](#)”, **Weishun Zhong**, Oles Shtanko, Ramis Movassagh, *arXiv:2401.06300, under review*.
5. “[Many-body Localized Hidden Generative Models](#)”, **Weishun Zhong**, Xun Gao, Susanne Yelin, Khadijeh Najafi, *arXiv: 2207.02346; Physical Review Research 6.4 (2024): 043041*.
6. “[A Theory of Weight Distribution-constrained Learning](#)”, **Weishun Zhong**, Ben Sorscher, Daniel D Lee, Haim Sompolinsky, *arXiv:2206.08933; NeurIPS 2022*
7. “[Quantifying Many-body Learning far from Equilibrium with Representation Learning](#)”, **Weishun Zhong***, Jacob M Gold*, Sarah Marzen, Jeremy L England, Nicole Yunger Halpern, *arXiv: 2001.03623; Scientific reports 11.1 (2021): 1-11*
8. “[Learning about Learning by Many-body Systems](#)”, **Weishun Zhong***, Jacob M Gold*, Sarah Marzen, Jeremy L England, Nicole Yunger Halpern, *arXiv:2004.03604; ICML workshop ML Interpretability for Scientific Discovery (2020)*
9. “[Non-equilibrium Statistical Mechanics of Continuous Attractors](#)”, **Weishun Zhong**, Zhiyue Lu, David J. Schwab, and Arvind Murugan, *arXiv: 1809.11167; Neural computation (2020) 32 (6)*

10. “A Closer Look at Disentangling in β -VAE”, Harshvardhan Sikka*, **Weishun Zhong***, Jun Yin, Cengiz Pehlevan, *arXiv:1912.05127; 53rd Asilomar Conference on Signals, Systems, and Computers (2019)*
11. “Associative Pattern Recognition in Macro-Molecular Self-Assembly”, **Weishun Zhong**, David J. Schwab, and Arvind Murugan, *arXiv: 1701.01769; J Stat Phys (2017) 167: 806*
12. “A Holographic c-Theorem for Schrödinger Spacetimes”, James T. Liu and **Weishun Zhong**, *arXiv: 1510.06975; JHEP 1512 (2015) 179*

AWARDS & HONORS

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| • Top Ten Highlights of the Year in Physics, <i>APS Physics Magazine</i> | 2025 |
| • Cosyne Travel Award, <i>Committee of Computational and Systems Neuroscience</i> | 2025 |
| • Eric and Wendy Schmidt Membership, <i>Simons Foundation</i> | 2024-2025 |
| • C.V. Starr Membership, <i>C.V. Starr Foundation</i> | 2023 |
| • NeurIPS Scholar Award, <i>Neural Information Processing Systems Foundation</i> | 2022 |
| • First-year Graduate Fellowship, <i>Massachusetts Institute of Technology</i> | 2017 |
| • Physical Sciences Division Tuition Award, <i>University of Chicago</i> | 2016 |
| • George Eugene Uhlenbeck Award, <i>University of Michigan</i> | 2016 |
| • Division of Particle and Fields Travel Award, <i>American Physical Society</i> | 2016 |
| • Division of Gravitational Physics Travel Award, <i>American Physical Society</i> | 2016 |
| • Otto Graf Scholarship, <i>University of Michigan</i> | 2015 |
| • James B. Angell Scholar, <i>University of Michigan</i> | 2015 |

GRANTS

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| • NSF ACCESS Discover Grant (PI), “Using Large Language Models to Understand the Cognitive Origin of Summarization and Abstraction”, <i>National Science Foundation, USA</i> | 2024-2026 |
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TEACHING EXPERIENCE

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|---|------------|
| • Teaching assistant for MIT (8.592) Statistical Physics in Biology | 2021 |
| • Teaching assistant for MIT (8.03) Physics III: Vibrations and Waves | 2020 |
| • Teaching assistant for MIT (8.02) Physics II: Electricity and Magnetism | 2019, 2022 |
| • Teaching assistant for MIT (8.01) Physics I: Classical Mechanics | 2018 |

SELECTED INVITED TALKS

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| • “Statistical Mechanics of Real Memories in Natural Language”,
<i>Statistical Physics Seminar</i> , University of Maryland – College Park | Feb.2026 |
| • “The Structure of Meaning: A Statistical Theory of Naturalistic Memory and Language”,
<i>Brain and Cognitive Sciences Seminar</i> , University of Rochester | Feb.2026 |
| • “Toward the Physics of Intelligence: From Neurons to Language”,
<i>Physics Colloquium</i> , University of Miami | Feb.2026 |
| • “Statistical Mechanics of Chunking – from Natural Language to Human Memory”,
<i>Statistical Physics Youth Communications Seminar Series (Online)</i> | Nov.2025 |
| • “Semantic Chunking and the Entropy of Natural Language”,
<i>Center for Brain Science Seminar</i> , Harvard University | Nov.2025 |
| • “Semantic Chunking and the Entropy of Natural Language”,
<i>Biophysics Seminar</i> , Princeton University | Oct.2025 |
| • “Synaptic Theory of Chunking in Working Memory”,
<i>Human Neurophysiology Seminar</i> , Cedars-Sinai Medical Center and Caltech | Sep.2025 |
| • “Random Tree Model for Meaningful Memory”,
<i>Simons Center for Systems Biology Seminar</i> , Institute for Advanced Study | May.2025 |
| • “Random Tree Model for Meaningful Memory”,
<i>Biophysics Seminar</i> , Princeton University | Mar.2025 |

- “Advantage of Quantum Neural Networks as Quantum Information Decoders”, Jan.2025
Quantum Meets Mathematics Seminar, Ohio State University and Harvard University
- “Learning in Feed-forward Neural Networks under Biological Constraints”, Apr.2023
Physics of Living Systems Seminar, Massachusetts Institute of Technology
- “Many-body Localized Hidden Generative Models”, Jan.2023
Center for Computational Quantum Physics Seminar, Flatiron Institute
- “A Theory of Learning with Constrained Weight-distribution”, Dec.2022
Simons Center for Systems Biology Seminar, Institute for Advanced Study
- “Weight-distribution Constrained Learning”, Jan.2022
Statistical Mechanics Seminar, Massachusetts Institute of Technology
- “Neural Network as a Many-Body System: from Interaction to Function”, Jan.2021
Shenzhen Institute of Advanced Technology Neuroscience Seminar, Chinese Academy of Sciences.
- “Learning about Learning by Many-Body Systems”, Sep.2020
Center for Astrophysics Seminar, Harvard University
- “Learning about Learning by Many-Body Systems”, Jul.2020
ML Interpretability for Scientific Discovery workshop, ICML
- “Learning about Learning by Many-Body Systems”, May.2020
Physics of Living Systems Seminar, Massachusetts Institute of Technology
- “A Closer Look at Disentangling in β -VAE”, Nov.2019
Asilomar Conference on Signals, Systems, and Computers, Asilomar
- “A Closer Look at Disentangling in β -VAE”, Oct.2019
Applied Math Seminar, Harvard University

SELECTED BROADER IMPACTS

- “[Highlights of the Year](#)”. Dec.2025
Featured at APS *Physics* magazine.
- “[Mathematical Model Reveals How Humans Store Narrative Memories Using ‘Random Trees’](#)”. Jul.2025
Featured at Science X *Medical Xpress* magazine.
- “[How We Remember Stories](#)”. Jun.2025
Featured at APS *Physics* magazine.
- “[IAS Scholars Discover a Universal Law of Memory](#)”. Jun.2025
Featured at Institute for Advanced Study News & Ideas.
- “[Investigating Natural and Artificial Intelligence: Q&A with Weishun Zhong](#)”. Sep.2023
Featured at Institute for Advanced Study News & Ideas.